

**1. (4 pontos) (Based on Kurlat (2020, Cap. 7))**

Consider the static problem of a representative household choosing consumption and labor to maximize the following utility function:

$$U = \ln(c) + b \frac{(l)^{1-\gamma}}{1-\gamma} \quad (1)$$

where  $c$  is consumption,  $l$  is leisure, and parameters  $b, \gamma > 0$ . The household pays a fraction  $\tau$  of its wages as taxes and receives a lump-sum transfer from the government of  $T$ .

- Derive the labor supply equation for the household, characterize it graphically, and discuss its properties.
- What is the effect of an increase in  $\tau$  on labor supply? Show the result mathematically and graphically, and discuss the intuition behind it.

**2. (6 pontos) Based on Kurlat (2020, Cap. 7)**

Consider the search model of Diamond, Mortensen and Pissarides (DMP). Let  $U$  be the number of unemployed workers,  $V$  the number of vacancies posted by firms, and  $E$  the number of employed workers. The labor force  $L = U + E$  is constant. The number of new matches formed each period is given by the matching function

$$m(V, U) = \mu V^\alpha U^{1-\alpha} \quad (2)$$

where  $\alpha \in (0, 1)$  and  $\mu > 0$ . Each period, a fraction  $s \in (0, 1)$  of employed workers loses their job exogenously. Define the unemployment rate  $u \equiv \frac{U}{L}$ , the vacancy rate  $v \equiv \frac{V}{L}$ , and market tightness  $\theta \equiv \frac{V}{U}$ .

- Write down the law of motion of employment,  $E_{t+1}$ , as a function of  $E_t$ ,  $V_t$  and  $U_t$ . Explain what each term represents. What does the parameter  $\mu$  measure, and how does a higher  $\mu$  change the law of motion?
- Impose steady state and derive the equation that equates job creation and job destruction. Use it to obtain the Beveridge curve of this economy. Draw it in  $(u, v)$  space and explain why it slopes the way it does.
- Define the job finding rate  $f \equiv \frac{m}{U}$  and the job filling rate  $q \equiv \frac{m}{V}$ . Write both as functions of  $\theta$  alone. Show that  $f = \theta q$ , and say whether each rate rises or falls with  $\theta$ .
- Suppose a single firm posts one extra vacancy. What happens to  $f$  and to  $q$ ? Which agents in the economy gain and which lose? Does the firm take these effects into account when it decides to post the vacancy?
- What is the effect of a fall in  $\mu$  on the Beveridge curve? What does this imply for the vacancy rate at a given unemployment rate? Explain how this differs from a movement along the curve.